Preprint

SUPPLEMENTARY MATERIAL

The SST Covariant Completion in Its Quasi-Static Limit

Stephen Nelson

Field Dynamics Research
stephen@fielddynamics.org

Abstract
The scalar-tensor transition (SST) is the four-term covariant action of Gravity Field Dynamics, with kinetic function $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$. We derive, exactly, what it gives in the quasi-static weak-field limit in which galaxies are observed. We read the action as the framework does: Φ is the Newtonian potential of the metric, the curvature term carries the tensor sector, and Φ's response is carried by F. Lemma 1 states that reading as an exact identity for the static Lagrangian. From it the curvature term's equation gives $\Psi = \Phi$, so $\gamma = 1$ at leading order, and the Φ-equation is $\nabla\cdot[\mu(|\nabla\Phi|/a_0)\nabla\Phi] = 4\pi G\rho$ with $\mu(x) = x/(x+1)$, equivalently the capacity form with $f(x) = (21x+13)/(x+1)$. Minimal coupling of matter gives the source. In spherical symmetry the field equation integrates exactly to $g_s = g_t^2/(a_0+g_t)$, whose forward solver and mass decoder are exact inverses. We derive the Newtonian and deep limits, the baryonic Tully–Fisher relation, the bound on the TetB share, the transition ring, the phantom density and well-posedness, and, from $\gamma = 1$, light deflection $4GM/(bc^2)$ in the Newtonian field and $2\pi\sqrt{GMa_0}/c^2$ in the deep field, with lensing mass equal to dynamical mass.

Keywords: scalar-tensor transition, covariant action, AQUAL, constitutive law, radial acceleration relation, Tully–Fisher, lensing, proof

Setting

The SST is the action [1, 2, 3]

$$S = \int d^4x\,\sqrt{-g}\,\Big[\frac{R}{16\pi G} - \frac{a_0^2}{8\pi G}\,F(y) - \tfrac{1}{4}e^{-2\Phi/c^2}F_{\mu\nu}F^{\mu\nu} + \mathcal{L}_{\text{matter}}\Big], \tag{1}$$

with $y = |\nabla\Phi|^2/a_0^2$ and $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$. F is the unique energy of the constitutive law $\mu(x) = x/(x+1)$, $F'(y) = \mu(\sqrt{y})$, $F(0) = 0$ [4]; μ is derived from the exchange symmetry of the two roots of the tetrad equation [5, 6]. We use the framework's reading of the action, as stated in the archive:

(i) Φ is the Newtonian potential of the metric, "not a new auxiliary field; the partition modifies its response, not its identity" [1], with the torsion share $1-\mu$ carried inside F [2].

(ii) The curvature term carries the tensor sector, so gravitational waves travel at c, and Φ's response is carried by F, the scalar sector "decoupling from the metric dynamics" [2]. Lemma 1 states this exactly.

(iii) Matter is minimally coupled to the metric, and $e^{-2\Phi/c^2} \to 1$ in the static weak field [7].

(iv) μ applies to the net field of a galaxy, not to each mass; a bound system much smaller than its host, such as the solar system, is pure TetA, that is, general relativity [8, 9].

Naming. "SST" names the action; the spherical, static relation $g_s = g_t^2/(a_0+g_t)$ is its spherical field equation [8]. The documentation calls the capacity function $f(x) = (21x+13)/(x+1)$ "the field equation" [10]; Theorem 2 shows that the two are one equation.

Contributions and prior work. The AQUAL field equation, its spherical reduction, the Newtonian and deep limits, the Tully–Fisher relation, well-posedness and the deep-field deflection are standard results for the simple interpolating function [14, 16, 15]. What this proof adds is their derivation from the SST's own terms, with the framework's reading stated as an exact identity (Lemma 1) and $\gamma = 1$ derived from it, together with the capacity form, the exact solver–decoder pair, the TetB-share bound and the transition ring, each checked by machine.

The Field Equation

Write the static weak-field metric as $g_{00} = -(1+2\Phi/c^2)$, $g_{ij} = (1-2\Psi/c^2)\delta_{ij}$.

Lemma 1 (the reading, exactly). *The static part of $\sqrt{-g}R/16\pi G$ at second order is, up to a divergence, $(8\pi G)^{-1}[(\nabla\Psi)^2 - 2\nabla\Phi\cdot\nabla\Psi]$, which on $\Psi = \Phi$ is the Newtonian energy $-(\nabla\Phi)^2/8\pi G = -(a_0^2/8\pi G)\,y$. Reading (ii) is the statement that the static Lagrangian is*

$$\mathcal{L}_{\text{st}} = \frac{(\nabla\Psi)^2 - 2\nabla\Phi\cdot\nabla\Psi}{8\pi G} - \frac{a_0^2}{8\pi G}\big[F(y) - y\big] - \rho\Phi, \tag{2}$$

the curvature term supplying the Newtonian part y of F. On $\Psi = \Phi$ this is $-(a_0^2/8\pi G)F(y) - \rho\Phi$.

The source term is minimal coupling: a mass m contributes $-mc^2\sqrt{-g_{00}} = -mc^2 - m\Phi + O(\Phi^2/c^2)$.

Proof. The second-order expansion of $\sqrt{-g}R$ for this metric is computed directly; its Euler–Lagrange equations are twice those of $(\nabla\Psi)^2 - 2\nabla\Phi\cdot\nabla\Psi$. The rest is substitution. □

Theorem 1. *Varying Eq. (2) with respect to Ψ gives $\nabla^2(\Phi-\Psi)=0$, so $\Psi=\Phi$ for fields that vanish at infinity: $\gamma=1$ at leading order. Varying with respect to Φ and setting $\Psi=\Phi$ gives*

$$\nabla\cdot\Big[\mu\Big(\frac{|\nabla\Phi|}{a_0}\Big)\nabla\Phi\Big] = 4\pi G\rho. \tag{3}$$

Proof. The Ψ-equation is $(8\pi G)^{-1}(-2\nabla^2\Psi + 2\nabla^2\Phi) = 0$. In the Φ-equation, the curvature term contributes $\nabla^2\Psi/4\pi G$ and the term $+(a_0^2/8\pi G)y$ contributes $-\nabla^2\Phi/4\pi G$; these cancel on $\Psi=\Phi$. For the rest, $\partial F/\partial(\nabla\Phi) = F'(y)\,2\nabla\Phi/a_0^2 = 2\mu\nabla\Phi/a_0^2$, and $(a_0^2/8\pi G)(2/a_0^2) = 1/(4\pi G)$. The Euler–Lagrange equation is $(4\pi G)^{-1}\nabla\cdot(\mu\nabla\Phi) - \rho = 0$, the Bekenstein–Milgrom form [14]. □

Theorem 2 (the capacity form). *With $\sigma(x) = (x-1)/(x+1)$, the capacity function $f(x) = 17 + 4\sigma(x) = (21x+13)/(x+1)$ takes the values $13, 17, 21$ at $x = 0, 1, \infty$, and $\mu = (f-13)/8$. So Eq. (3) is $\nabla\cdot[\frac{1}{8}(f-13)\nabla\Phi] = 4\pi G\rho$, the form of Paper VIII and the documentation.*

Proof. $17(x+1) + 4(x-1) = 21x + 13$, and $(21x + 13 - 13x - 13)/8 = x$. □

The Spherical Field Equation

Theorem 3. *In spherical symmetry, with $g_t = |\Phi'(r)|$ and $g_s = GM(<r)/r^2$, Eq. (3) integrates exactly to the conserved flux $r^2\mu(g_t/a_0)\,g_t = GM(<r)$, that is,*

$$g_s = \frac{g_t^2}{a_0 + g_t}, \qquad g_t = \tfrac{1}{2}\Big[g_s + \sqrt{g_s^2 + 4g_s a_0}\Big], \tag{4}$$

the second being the unique positive root. With $g_t = v^2/r$, the mass decoder

$$M_{\rm dec}(r) = \frac{rv^4}{G(a_0 r + v^2)} \tag{5}$$

is the exact inverse of the forward solver.

Proof. For $\Phi = \Phi(r)$, $\nabla\cdot(\mu\nabla\Phi) = r^{-2}(r^2\mu\Phi')'$, and $4\pi r^2\rho = M'(r)$, so the divergence theorem gives the flux with no curl term. Substituting $\mu = x/(x+1)$, $x = g_t/a_0$, gives the quadratic $g_t^2 - g_s g_t - g_s a_0 = 0$, whose other root is negative. The decoder is $r^2 g_s/G$ at $g_t = v^2/r$. □

This is the relation fitted to the SPARC rotation curves with no parameter in μ [7], solved forward in Paper X [11] and inverted in the decoding-mass letter [12].

Limits and Scales

Theorem 4. *(i) As $g_s \to \infty$, $g_t = g_s + a_0 - a_0^2/g_s + O(g_s^{-2})$: Newtonian. (ii) As $g_s \to 0$, $g_t = \sqrt{g_s a_0} + g_s/2 + O(g_s^{3/2})$. (iii) For a bounded, isolated mass M, $v^4 \to GMa_0$ exactly as $r \to \infty$: the baryonic Tully–Fisher relation (an external field modifies this, as in AQUAL). (iv) The TetB share of the field is $(1-\mu)g_t = a_0\mu(g_t/a_0) < a_0$. (v) The ring $x = 1$, where $\mu = \frac{1}{2}$, lies where $g_s = a_0/2$, at $r_1 = \sqrt{2GM/a_0}$; at $r = \sqrt{GM/a_0}$, where $g_s = a_0$, the field is at $x = \varphi = (1+\sqrt{5})/2$ with $\mu = 1/\varphi$. (vi) Outside a point mass the phantom density $\rho_{\rm ph} = (4\pi G r^2)^{-1}\,d(r^2 g_t)/dr$ has leading term $\sqrt{GMa_0}/(4\pi G r^2)$.*

Proof. (i), (ii): expansions of the forward solver. (iii): $v^4 = g_t^2 r^2$ with $g_s = GM/r^2$, and the limit of the solver. (iv): $g_t/(1+x) = a_0 x/(1+x)$. (v): $x = 1$ gives $g_s = 1/(1+1)\cdot a_0$; $g_s = a_0$ gives $x^2 = x + 1$. (vi): $r^2 g_s = GM$ is constant outside the mass, so the residual of the Newtonian flux is $r^2 g_t$, which grows as $\sqrt{GMa_0}\,r$. □

Theorem 5 (well-posedness). *The energy density $F(|p|^2/a_0^2)$ is strictly convex in $p = \nabla\Phi$ for $p \neq 0$, with Hessian eigenvalues 2μ across and $2(x\mu)'$ along the field, whose ratio $(x\mu)'/\mu = 1 + (1-\mu)$ lies in $[1, 2]$. So Eq. (3) is elliptic away from $\nabla\Phi = 0$, and its solution is unique for given boundary data or for fields decaying at infinity. Existence for AQUAL is due to Bekenstein and Milgrom [14].*

Proof. The energy-functional proof [4]; the eigenvalues are $2x/(x+1)$ and $2x(x+2)/(x+1)^2$. □

The Single Potential

Theorem 6. *By Theorem 1 the weak-field metric carries one potential in both components, $g_{00} = -(1 + 2\Phi/c^2)$, $g_{ij} = (1 - 2\Phi/c^2)\delta_{ij}$, so the lensing potential is $\int(\Phi+\Psi)\,dz = 2\int\Phi\,dz$: light is deflected by the same Φ that sets the dynamics. In the Born approximation the deflection at impact parameter b is*

$\theta = (2/c^2)\int g_\perp\, dz$. *A Newtonian point mass gives* $4GM/(bc^2)$*; the deep field* $g_t = \sqrt{GMa_0}/r$ *gives the constant* $2\pi\sqrt{GMa_0}/c^2$*; and for a spherical lens the lensing mass equals the dynamical mass* $M_{\rm dyn}(r) = g_t r^2/G$*. In a galaxy the factor* $e^{-2\Phi/c^2}$ *differs from* 1 *by about* $2|\Phi|/c^2 \sim 10^{-6}$*, so the fourth term does not enter the static dynamics.*

Proof. The deflection is $((1+\gamma)/c^2)\int \nabla_\perp\Phi\, dz$ with $\nabla_\perp\Phi = g_t\, b/r$; the integrals $\int b\, r^{-3} dz = 2/b$ and $\int b\, r^{-2} dz = \pi$ give the two values. A general-relativistic lens with mass profile $M_{\rm dyn}$ has $g = GM_{\rm dyn}/r^2 = g_t$, the same integrand. □

This is Letter 3's statement that $\gamma = 1$ "because the single potential Φ governs both metric components", and Paper I's single-standard postulate [13], here derived from the curvature term's Ψ-equation. The deep-field deflection is Milgrom's [16].

Remark (the reading in (ii)). The static weak-field part of $R/16\pi G$ is $(8\pi G)^{-1}[(\nabla\Psi)^2 - 2\nabla\Phi\cdot\nabla\Psi]$, which on $\Psi = \Phi$ is the Newtonian energy $-(\nabla\Phi)^2/8\pi G$. If F were added to it in full, rather than $F - y$ as in Lemma 1, the equations would give $\Psi = \Phi$ and $\nabla\cdot[(1+\mu)\nabla\Phi] = 4\pi G\rho$. The SST's field equation, Eq. (3), is the one in which Φ's response is carried by F alone, as the archive states; Lemma 1 is that statement made exact.

Scope

Theorems 1–6 are exact consequences of the SST, read as in Lemma 1, at leading order in the quasi-static weak field, and Theorems 3–4 are what the SPARC rotation curves, the Tully–Fisher relation and the mass decoder test. The limit uses the static frame, in which $|\nabla\Phi|^2$ is defined; as a spacetime scalar, y requires Φ to be a scalar field, and its covariant form for time-dependent and strong fields is given in the companion proof (`proofs/sst-covariant-time-leg/`): $y = c^4 a_\mu a^\mu/a_0^2$, with a the acceleration of the tetrad's time leg, and Φ the lapse. The archive records what else remains open [8]: the metric field equations beyond leading order, including the stress–energy of the F sector; $\mathcal{L}_{\rm matter}$ beyond its weak-field form; the documents describe the fourth term differently (disformal and needed for lensing; conformal with a cosmological role; an auxiliary vector), and the dependence of the fine-structure constant on the potential that it implies is not discussed; and the match from the locked bimetric theory to μ is open. At the radius $\sqrt{GM/a_0}$ quoted as the processing boundary in Paper I, the Newtonian acceleration equals a_0, and there $x = \varphi$ and $\mu = 1/\varphi$; the ring $\mu = \frac{1}{2}$ is at $\sqrt{2GM/a_0}$ (Theorem 4(v)).

References

[1] S. Nelson, *Gravity Field Dynamics: Field Closure of the Tetrad Frame*, Letter 3 (2026).

[2] S. Nelson, *Covariant completion*, supplementary note to Letter 3 (2026).

[3] S. Nelson, *A Covariant Theory Without Dark Matter*, Paper II (2026), doi:10.5281/zenodo.18360354.

[4] S. Nelson, *The energy functional of the constitutive law* (2026), `proofs/aqual-energy-functional/`.

[5] S. Nelson, *The Other Half of Gravity: Torsion and the Two-Chamber Structure of the Tetrad Frame*, Letter 2 (2026).

[6] S. Nelson, *The constitutive law from exchange symmetry* (2026), `proofs/constitutive-law-exchange-symmetry/`.

[7] S. Nelson, *The Radial Acceleration Relation from a Zero-Parameter Covariant Action* (2026).

[8] S. Nelson, *Gravity Field Dynamics — Coles Notes* (2026), §§3, 5.4, 12.

[9] S. Nelson, *Einstein's equation as the TetA partition* (2026).

[10] S. Nelson, *Documentation: The Field Equation; Covariant Completion* (2026), `docs/02-gravitation/`.

[11] S. Nelson, *One Equation, Two Scales: Tully–Fisher Evolution and the Hubble Constant*, Paper X (2026).

[12] S. Nelson, *Decoding Mass from Observed Rotation Curves* (2026), doi:10.5281/zenodo.18721752.

[13] S. Nelson, *The Gravitational Constitutive Law*, Paper I (2026), doi:10.5281/zenodo.18332595.

[14] J. Bekenstein and M. Milgrom, "Does the missing mass problem signal the breakdown of Newtonian gravity?", *Astrophys. J.* **286** (1984) 7.

[15] B. Famaey and S. S. McGaugh, "Modified Newtonian Dynamics (MOND): observational phenomenology and relativistic extensions", *Living Rev. Relativ.* **15** (2012) 10.

[16] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis", *Astrophys. J.* **270** (1983) 365.